

Strategic Search in Patenting

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Abstract

I develop a three-player game-theoretic model of patenting. An *applicant* chooses how extensively to search for prior art, determining the information brought into the patenting process and the claimed patent breadth. An *examiner* conducts additional search and might narrow the claimed breadth. A *challenger* conducts additional search and might challenge the patent. The public record from earlier search affects the likelihood of challenge by affecting the informativeness of later search. My equilibrium characterization yields comparative statics for how search and breadth respond to challenger filing costs, procedural standards, and examiner capacity. These results generate testable implications for patent breadth, grant outcomes, and post-grant challenge behavior.

Keywords: patent breadth; strategic search; information acquisition; patent prosecution; examiner search; post-grant challenge; patent quality

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1 Introduction

The U.S. patent system is meant to reward inventive activity while limiting patent rights to the inventor’s actual contribution. Overly narrow rights can diminish incentives to innovate, while overly broad rights can distort competition, deter follow-on invention, and create uncertainty for rivals operating near existing technologies. The economic problem is therefore whether the patent system produces rights with appropriate breadth.

A patent’s breadth is determined by searches for prior art by the applicant, the patent examiner, and later challengers. Before filing, applicants may search for prior art and therefore how aggressively to pursue broad claims. After filing, examiners conduct additional search, evaluate novelty and non-obviousness, and may narrow the application. After grant, competitors or other interested parties may invest in additional search and decide whether to challenge the patent in court or through administrative review.

Existing research analyzes different parts of the strategic search problem in patenting, but typically studies them in isolation. For example, Atal and Bar show applicants may have little incentive to search when search can reduce the probability of obtaining a patent, though applicant search can rise with examiner effort, patenting fees, and R&D costs [2]. Langinier and Marcoul show applicant revelation and examiner scrutiny are jointly determined, with examiner screening intensity responding to the information transmitted by the applicant [21]. Their later work shows monetary incentives and career concerns can intensify examiner search effort, underscoring that search incentives are endogenous and policy-sensitive on both sides of the patenting process [20].

I develop a three-player model providing a unified account of how strategic search by an applicant, examiner, and challenger jointly determines patent breadth. Applicant search determines how much information enters prosecution and how broad the application can plausibly be. Examiner search determines how much claimed breadth survives into the granted patent. The prosecution record shapes future challengers’ beliefs about whether additional search and filing are worthwhile.

Focusing on equilibrium breadth rather than the binary notion of validity is important because a patent’s economic effects often depend on how far its claims extend beyond the invention’s legally defensible core. A patent may be granted, licensed, asserted, and influence rivals’ behavior long before validity is fully tested. Empirical work on claim structure and examination supports this emphasis: examination tends to narrow claims, broader initial claims are associated with different

grant and prosecution outcomes, and examiner-driven variation in patent rights has meaningful downstream effects on value, litigation, and innovation [28, 10].

My model yields comparative statics characterizing how applicant search, examiner scrutiny, challenger search, filing behavior, and equilibrium breadth respond to changes in institutional features. For example, a change in the cost of challenging patents affects the expected payoff from making broad claims in the first place. If challenges become cheaper or more credible, applicants may have less incentive to pursue claims likely to be narrowed later and more incentive to search earlier to understand the limits of valid protection. Examiners may also respond differently when the downstream environment changes. The comparative statics also speak to examiner capacity, search requirements, and others policies that alter incentives to acquire information at different stages of the patenting process.

A natural application of the model is the reduction in challenger filing costs associated with Patent Trial and Appeal Board (PTAB) review. The America Invents Act (AIA) created new administrative routes for challenging issued patents, including post-grant review and inter partes review. These changes to the patenting process, alongside other policy shifts, create opportunities for empirical investigation.

Recent research suggests changes to downstream challenge institutions can shape applicant and examiner incentives before rights are granted. Schankerman and Schuett study screening for weak patents in a theoretical framework with endogenous challenge and show courts, even if perfect, cannot fully solve the screening problem because many weak patents are never challenged; they also show PTAB's welfare effect is theoretically ambiguous once indirect strategic effects are included [33]. Empirical evidence from Martinelli and Mazzei suggests post-grant review can generate upstream discipline: after the AIA, applicants in affected settings narrow patent breadth more aggressively, and examiners become tougher when applicants do not do so themselves [29]. Related empirical work shows examiner time constraints affect scrutiny and grant behavior, and greater ex ante examination effort can improve the validity of issued patents [12, 11].

By recognizing patent breadth as the equilibrium outcome of a multi-stage strategic search process, the model provides a framework for studying how pre-grant and post-grant institutions jointly determine the breadth of patent rights. Applicants do not choose claim breadth in isolation, examiners do not screen applications in a vacuum, and challengers do not evaluate patents independently of the information generated during prosecution. Structural changes at any stage can affect behavior

at all stages, because each stage shapes (and anticipates) later ones.

The paper proceeds as follows. Section 2 describes the U.S. patenting process, while Section 3 situates the paper within the patenting literature. Sections 4 and 5 respectively develop the model and characterize equilibrium behavior. Section 6 derives comparative statics with respect to challenge costs and other policy parameters. Section 7 discusses empirical implications, with particular attention to post-grant review and patent breadth. Section 8 concludes.

2 The U.S. Patenting Process

A patent application begins with a specification and claims describing the invention and defining the protection sought. The application must adequately disclose the invention, and later amendments must remain supported by the original filing; new matter cannot be added after filing.

The application enters *prosecution* after it is filed. Applicants may submit prior-art references through an information disclosure statement, respond to examiner objections with arguments and amendments, provide supporting evidence, and participate in examiner interviews that become part of the record. The Patent Office evaluates the claims against the relevant legal standards, then it decides whether the proposed breadth should be allowed, narrowed, or rejected [43, 39, 42].

Applicants and their representatives owe a duty of candor and good faith that requires them to disclose known information material to patentability, but they have no general duty to conduct exhaustive prior-art searches. That is, applicants must reveal what they have learned about prior art, but they are not required to learn anything. Consequently, the applicant-generated information entering prosecution is endogenous [40, 39].

The distinction between disclosure and search by applicants matters, because applicant search is the first institutional margin shaping patent breadth. The breadth an applicant can plausibly pursue depends on the prior art already known to the applicant and placed into the prosecution record. When applicants search more extensively, they may uncover constraining prior art before examination begins. When they search less, the examiner may inherit a thinner record and broader proposed claims. Applicant search therefore affects both the information available during prosecution and the breadth carried into examination.

Examiner search is the next institutional margin shaping patent breadth. After reviewing the specification and claims, the examiner searches the prior art and evaluates patentability in light of

the record. The examiner must consider prior art cited by the applicant or uncovered independently, and it should reject claims the prior art and record show are unlikely to be patentable. Examiner search and review therefore help determine how much of the breadth carried into prosecution survives into an allowed patent [41].

Finally, third parties may challenge issued claims through litigation or through administrative proceedings before the Patent Trial and Appeal Board. Post-grant review begins with a third-party petition within nine months of grant and may raise a broad set of invalidity grounds. Inter partes review allows review of one or more claims on prior-art grounds under Sections 102 or 103. These procedures make post-grant search and challenge part of the broader institutional environment in which patent breadth is determined [38].

The sequential production of information across stages is the key institutional feature shaping patent breadth. Applicants generate and disclose information before and during prosecution. Examiners add their own search and evaluate the claims in light of the combined record. Potential challengers observe the issued patent and prosecution record before deciding whether additional search and challenge are worthwhile.

3 Literature Review

Because this paper treats patent breadth as the central outcome of the patenting process, it first relates to the literature on patent breadth’s economic consequences. Classic work studies the tradeoff between patent breadth and innovation incentives, showing broader rights can increase rewards to inventors while also imposing distortions on rivals and follow-on innovators [19, 35, 13]. Empirical work likewise connects variation in patent breadth and informational content to private value of patents. Broader patents are associated with higher firm valuations, patent citations are informative about market value of firms, and procedural indicators such as patent family size and opposition outcomes are related to the value of patent rights [24, 14, 15]. More recent work moves closer to the object studied here, by measuring patent breadth from claims data; examination tends to narrow claims, while examiner-driven variation in patent rights affects value, litigation, and innovation [28, 10]. This literature establishes the economic relevance of patent breadth and motivates treating breadth as an outcome of the patenting process rather than as a fixed feature of an invention. The remaining literatures speak to the three stages through which that outcome is produced: applicant

information acquisition, examiner search and screening, and downstream challenge.

A second literature studies how applicants acquire and disclose information during prosecution. Atal and Bar show applicants may have weak incentives to search for prior art when search can reduce the probability of obtaining a patent, although those incentives rise with examiner effort, patenting fees, and R&D costs [2]. Langinier and Marcoul extend this logic by showing applicant disclosure and examiner scrutiny can be jointly determined, with examiner behavior responding to the information applicants reveal [21]. Empirical work on citations in prosecution reinforces the importance of distinguishing applicant-generated information from the total body of prior art. Examiners add a substantial share of citations in issued patents, applicant-submitted citations often are not central to actual rejections, and examiner citations appear more predictive of patent value than applicant citations [1, 7, 16]. Legal scholarship on inequitable conduct further highlights the distinction between acquiring information and withholding information already known, while doctrinal changes in *Exergen* and *Therasense* altered the conditions under which withholding can generate liability [6, 8, 36]. Together, this work motivates treating the applicant-generated record as a distinct endogenous input into prosecution.

A third literature studies examiner search and screening, emphasizing heterogeneity in effort and the institutional determinants of examination quality. Early empirical work shows examiners are heterogeneous and examination outcomes vary systematically with examiner characteristics rather than reflecting a uniform screening technology [5, 22]. Theoretical work frames patent examination as an incentive problem in which examiners must exert effort and truthfully reveal what they find, implying compensation, auditing, and career concerns can affect screening quality [34, 20]. More recent empirical work shows institutional constraints matter in practice. Reducing available review time lowers scrutiny and increases grants on the margin, greater examination resources can improve validity in high-stakes settings, and heavier workloads are associated with grant bias and lower examination quality [12, 11, 18]. This literature demonstrates examiner screening is endogenous and policy-sensitive, while also highlighting the importance of examination for the breadth and quality of issued patent rights.

A fourth literature studies patent validity and downstream challenge, including how post-grant review affects incentives before a patent is granted. Lemley and Shapiro characterize patents as probabilistic rights rather than determinate exclusion rights, while Farrell and Shapiro show how uncertainty about patent validity affects licensing and welfare [23, 9]. Schankerman and Schuett

bring examination and challenge together in a model of patent screening with endogenous court challenges, showing courts cannot fully solve the screening problem because many weak patents are never challenged [33]. On the institutional side, the America Invents Act created new administrative challenge routes, especially inter partes review (IPR) and post-grant review, lowering the cost of contesting some issued patents. Early evidence documents the practical importance of IPR, while later work uses PTAB outcomes to study the determinants of patent quality more directly [26, 27]. Most closely related to the present paper, Martinelli and Mazzei show post-grant review can generate upstream effects: after the AIA, applicants narrowed patent breadth more aggressively and examiners became tougher when applicants did not do so [29]. This literature demonstrates post-grant institutions can affect incentives before rights are granted.

Taken together, these literatures show applicant-side information production, examiner screening, and downstream challenge each affect the breadth and quality of patent rights. Applicant-side work explains why incentives to search for and reveal prior art may be weak or distorted [2, 21]. Examiner-side work shows scrutiny responds to office incentives, resources, and institutional design [34, 12, 11]. Challenge-side work explains why downstream review is selective and why post-grant institutions can affect behavior during prosecution [33, 29]. This paper builds on these insights by studying patent breadth as the outcome of strategic search across the patenting process. Rather than treating pre-grant search, examination, and post-grant review as separate margins, the framework emphasizes how incentives to acquire information at one stage can shape behavior at later stages, and how changes in post-grant institutions can influence claim breadth and examination before grant.

4 The Model Structure

This section presents a finite-horizon, three-player dynamic game comprised of seven steps grouped into the *applicant stage*, the *examiner stage*, and the *challenger stage*. The applicant begins with initial information and chooses whether to search further; the resulting set of knowledge determines the breadth carried into examination. The examiner observes the submitted record, chooses whether to search further, and truncates breadth according to the information in the record. The applicant then decides whether to accept or abandon the resulting patent. The potential challenger observes the granted breadth and prosecution record, chooses whether to search further, then decides whether to file a challenge. Challenge success is determined by the patent’s true overbreadth.

4.1 The Strategic Environment

Several parameters in the game are common knowledge. The patent's quality $q \in Q \subset \mathbb{R}_+$ acts as a scaling object on the value of patent breadth to the applicant and challenger. This formulation differs from those in which quality is an endogenous valuation object [31, 25, 32]. The parameters λ_A, λ_E , and λ_C affect each player's marginal cost of search.

Other parameters are not common knowledge. No player observes the patent's legally valid breadth $b^* \in B \subset \mathbb{R}_+$, which Nature draws from a commonly known distribution F_0 . The applicant and challenger each privately observe their respective legal costs $K_A \sim G_A$ and $K_C \sim G_C$, which are incurred only if a challenge occurs after grant. Finally, the applicant privately learns its initial information X_0 , which represents research conducted before the beginning of the game.

The game's timing is as follows.

1. Nature draws b^* , K_C , K_A , and X_0 .
2. The applicant chooses search effort s_A after observing X_0 , which generates information X_1 .
3. The examiner chooses search effort s_E after observing X_1 , which generates information X_2 and granted breadth b_E .
4. The applicant chooses whether to accept the granted breadth or abandon the patenting process, after observing X_2 and b_E .
5. Conditional on grant, the challenger chooses search effort s_C after observing X_2 and b_E , which generates information X_3 .
6. The challenger decides whether to file a challenge, after observing X_3 .
7. If a challenge is filed, the true breadth b^* is revealed and challenge success is probabilistically determined by the realized overbreadth $b_E - b^*$.

Search arises in the model because no player knows the legally valid breadth, but patent value depends on granted breadth and a challenge's value and likely success depend on the relationship between granted breadth and legally valid breadth. Each player's search decision is based on prior information, and anticipates future search decisions.

My model abstracts from the applicant's choice of which discovered references to place into the prosecution record, which seems natural if the expected penalty from concealing known material information is large.¹ In U.S. patent law, the relevant deterrent is inequitable conduct: if an applicant knowingly withholds material prior art with deceptive intent, the resulting patent may be rendered unenforceable, and the attorneys involved may face professional or legal consequences. If the legal risk is large enough, the applicant's most relevant strategic margin is not whether to disclose discovered prior art, but whether to acquire that information in the first place.

4.2 The Player's Payoffs and Search Costs

The players differ in what they value from the patenting process. If a patent granted at breadth b_E survives any post-grant challenge, the applicant receives value

$$V_A(b_E, q) = v(q)u(b_E),$$

where $v'(q) > 0$, $u'(b_E) > 0$, and $u''(b_E) < 0$. Patent quality scales the value of the right, while breadth has positive but diminishing marginal value. The applicant's return is evaluated net of any search costs and legal costs it incurs.

If a patent granted at breadth b_E is successfully challenged, then the challenger receives value

$$V_C(b_E, q) = v(q)u(b_E),$$

using the same basic value structure as the applicant. This symmetry is not essential, but it captures the idea that broader and higher-quality patents are more valuable both to hold and to remove as competitive constraints. The challenger's return is evaluated net of any search costs and legal costs it incurs.

The examiner does not value patent breadth directly, but dislikes granting claims exceeding the legally valid breadth. Regardless of acceptance or challenge, the examiner's loss is

$$V_E(b_E - b^*),$$

¹Including a disclosure stage would make the model richer, but it would also combine the search mechanism with a separate concealment problem, significantly decreasing the model's tractability.

where $V_E(0) = 0$, $V'_E(b_E - b^*) > 0$, and $V''_E(b_E - b^*) > 0$. The examiner's return is evaluated net of any search costs it incurs.²

Each player $i \in \{A, E, C\}$ incurs a search cost $c_i(s_i; \lambda_i)$, where λ_i is a search-cost parameter. Search costs are increasing in s_i , and c_i has increasing differences in (s_i, λ_i) . Thus, a higher λ_i raises the incremental cost of greater search. For differentiable cost functions, this condition is equivalent to

$$\frac{\partial^2 c_i(s_i; \lambda_i)}{\partial s_i \partial \lambda_i} \geq 0.$$

The increasing-differences formulation does not require differentiability and is therefore convenient given the discrete search choices used in the model.

4.3 Beliefs and the Truncation-Draw Information Structure

Search generates observations that each can be interpreted as a piece of prior art placing an upper bound on the patent's legally valid breadth b^* . Formally, conditional on b^* , each search process $t \in \{0, A, E, C\}$ generates a finite set of constraining draws

$$X_t = \{x_1, \dots, x_{s_t}\}, \quad x_i \geq b^*.$$

Each search outcome is combined with prior information to generate each information state

$$X_1 \equiv X_0 \cup X_A \quad X_2 \equiv X_1 \cup X_E \quad X_3 \equiv X_2 \cup X_C$$

For each realized record X_j , $j \in \{0, 1, 2, 3\}$ define

$$m_j(X_j) \equiv \min(X_j), \quad \text{and} \quad R(X_j) \equiv |X_j|,$$

I write these objects simply as m_j and R_j . The statistic m_j is the tightest observed upper bound on b^* , while R_j is the size of the accumulated record. The model assumes players condition on the record only through these two statistics, so beliefs about b^* are summarized by the probability

²The examiner payoff is intentionally modeled as depending only on expected overbreadth. This keeps the examiner's role focused on ex ante screening. Allowing examiner utility to depend on later challenge or invalidation outcomes, for example through reputation or career concerns, would add another incentive to search but would not change the core mechanism emphasized here.

distribution

$$F(\cdot \mid m_j, R_j),$$

with posterior mean

$$\mu(m_j, R_j) \equiv E[b^* \mid m_j, R_j].$$

The two statistics affect beliefs through distinct channels. The minimum m creates a boundary effect: because each constraining draw provides an upper bound on b^* , posterior support is contained in $[0, m]$, and a lower minimum shifts beliefs toward lower legally valid breadth. For any fixed granted breadth, this shift tends to increase expected overbreadth. Record size R instead creates a screening effect: holding m fixed, a larger record makes the observed minimum more informative and shifts beliefs toward greater concentration near m , tending to reduce expected overbreadth. The following assumptions formalize these posterior properties.

Assumption 1 *Support restriction:* For every admissible (m, R) ,

$$\text{supp}(F(\cdot \mid m, R)) \subseteq [0, m].$$

Assumption 2 *Minimum monotonicity:* Holding R fixed, a lower minimum m shifts posterior beliefs toward lower values of the true legally valid breadth b^* in the sense of first-order stochastic dominance. Formally, for any $m' < m$,

$$F(b \mid m', R) \geq F(b \mid m, R) \quad \text{for all } b \in B.$$

Assumption 3 *Record-size monotonicity:* Holding m fixed, a larger record shifts posterior beliefs toward greater concentration near m . Formally, if $R' > R$, then

$$\mu(m, R') \geq \mu(m, R), \quad \text{with} \quad \lim_{R \rightarrow \infty} \mu(m, R) = m.$$

Much of the model's comparative-static logic follows from the interaction between these boundary and screening effects. Figure 1 illustrates these two belief channels.

The belief structure is similar at each stage, but with different realized records. The applicant begins from (m_0, R_0) , their search produces (m_1, R_1) , and the breadth entering examination is m_1 .

Examiner search updates the record to (m_2, R_2) , and the granted breadth $b_E = m_2$. Conditional on grant, challenger search updates the prosecution record to (m_3, R_3) . At every stage, posterior beliefs over b^* are summarized by the current pair (m_j, R_j) .

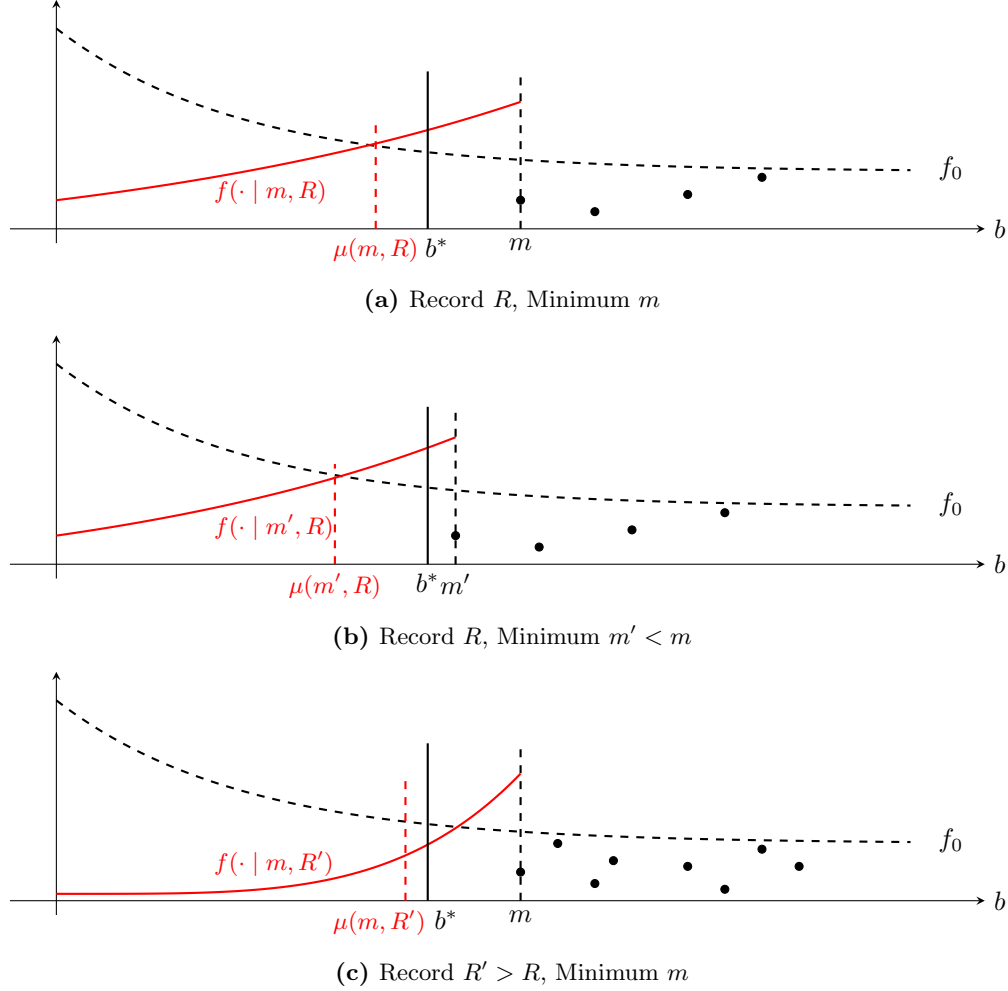


Figure 1: Truncation-draw belief updating. Each realized draw is a constraining observation lying weakly above the true legally valid breadth b^* . Panel (a) shows a baseline record with minimum draw m and posterior density $f(\cdot | m, R)$. Panel (b) keeps record size fixed but lowers the minimum to $m' < m$, which lowers the upper bound on feasible legally valid breadth and shifts the posterior mean downward. Panel (c) holds the minimum fixed at m but increases the number of observed draws to $R' > R$. The larger record makes the observed minimum more informative and shifts posterior beliefs toward m .

I use the truncation-draw structure because it more realistically captures the stochastic nature of prior-art discovery. Prior-art search does not proceed deterministically toward a known legal boundary. When a player searches, they do not know whether the search will uncover a new constraint

that materially changes the patent’s legally supportable breadth. Using a deterministic search technology would mechanically translate greater effort into narrower breadth or more precise beliefs, while using a direct-revelation technology would make search a probabilistic discovery of b^* itself. By contrast, the truncation-draw structure reflects the institutional setting in which search uncovers pieces of prior art that impose upper bounds on valid breadth. Additional search may therefore either uncover a more constraining reference and lower the observed minimum or expand the record without changing that minimum.

Search is discrete for both substantive and technical reasons. Substantively, prior-art search is often organized in discrete increments such as search queries, attorney hours, database searches, examiner time allocations, or challenger investigation steps. Technically, a finite search space ensures each player’s optimization problem admits a solution without imposing stronger smoothness, concavity, or single-crossing assumptions.

Because a marginal increase in search discontinuously increases the record and can discontinuously decrease the minimum, player’s continuation values need not be differentiable and their best-responses may be set-valued. The equilibrium analysis therefore uses monotone comparative statics rather than first-order conditions. The Topkis-style arguments below characterize monotone selections from these best-response correspondences without requiring differentiability of the player’s payoff functions [37, 30].

5 Equilibrium Characterization

I solve the game via backward induction. I first characterize the challenger’s filing and search decisions, then derive the applicant’s accept-or-abandon rule, the examiner’s search decision, and the applicant’s initial search decision. The resulting strategies form a subgame-perfect Nash equilibrium, which is suitable despite the applicant’s defense cost K_A and the challenger’s filing cost K_C being privately observed. These costs do not enter other player’s payoff functions directly. They affect continuation play only through the actions they induce, which earlier-moving players evaluate using the known cost distributions G_A and G_C .

5.1 The Challenger Stage

Conditional on the applicant accepting the granted patent, the challenger observes the inherited prosecution record (m_2, R_2) and the granted breadth $b_E \equiv m_2$. The challenger chooses whether to acquire additional information through search, then observes the record (m_3, R_3) and decides whether to file a challenge. Consequently, I characterize the filing rule first to define the value of search.

5.1.1 Challenger Filing

The probability a challenge succeeds is

$$p_C(b^*, b_E) = \rho(b_E - b^*),$$

where $\rho : \mathbb{R}_+ \rightarrow [0, 1]$ is weakly increasing in realized overbreadth. For a realized post-search information state (m_3, R_3) , the challenger's posterior-expected probability of success is

$$\bar{p}_C(b_E; m_3, R_3) \equiv \mathbb{E}[\rho(b_E - b^*) \mid m_3, R_3].$$

The challenger files if and only if the expected payoff from filing is non-negative, yielding the threshold rule

$$K_C \leq \hat{K}_C(b_E; m_3, R_3) \equiv V_C(b_E, q) \bar{p}_C(b_E; m_3, R_3).$$

Thus, conditional on the post-search state, filing is characterized by a cutoff in the challenger's privately observed challenge cost. If a challenge is filed, its realized probability of success is $\rho(b_E - b^*)$.

5.1.2 Challenger Search

Before observing the outcome of additional search, the challenger chooses search intensity

$$s_C \in S = \{0, 1, \dots, \bar{s}\},$$

with cost $c_C(s_C; \lambda_C)$, where search costs are increasing and convex in effort. Search generates additional constraining draws

$$X_C(s_C) = \{x_1, \dots, x_{s_C}\},$$

which are combined with the inherited prosecution record X_2 to produce

$$m_3(s_C) = \min(X_2 \cup X_C(s_C)), \quad R_3(s_C) = |X_2 \cup X_C(s_C)|.$$

The challenger chooses s_C to maximize

$$U_C(s_C; \theta_2) = \mathbb{E}[\max\{0, V_C(b_E, q)\bar{p}_C(b_E; m_3(s_C), R_3(s_C)) - K_C\} \mid m_2, R_2] - c_C(s_C; \lambda_C),$$

where

$$\theta_2 = (q, b_E, m_2, R_2, \lambda_C, K_C)$$

denotes the challenger's state. Search is valuable because it changes the distribution of post-search states (m_3, R_3) , and therefore the expected option value of filing.

5.1.3 Existence of a Challenger Search Best Response

The challenger's search problem admits a solution because, under the maintained assumptions, $U_C(s_C; \theta_2)$ is well defined and real valued for every $s_C \in S$, and S is finite. Therefore,

$$BR_C(\theta_2) \equiv \arg \max_{s_C \in S} U_C(s_C; \theta_2)$$

is nonempty. Because the best-response correspondence may contain multiple elements, I define its lower and upper selections as

$$s_C^L(\theta_2) \equiv \min BR_C(\theta_2) \quad \text{and} \quad s_C^H(\theta_2) \equiv \max BR_C(\theta_2).$$

When I refer to equilibrium challenger search below, s_C^* denotes one such selection.

5.2 The Applicant Accept or Abandon Decision

The applicant's accept-or-abandon decision anticipates the challenger stage. After examiner search, the applicant observes the prosecution record (m_2, R_2) and granted breadth $b_E \equiv m_2$, then decides whether to accept the patent or abandon the process.

Two probabilities summarize downstream challenge risk. Because K_C is privately observed by the challenger but drawn from the known distribution G_C , earlier players perceive challenge proba-

bilistically. From the applicant's perspective, K_C is distributed according to G_C , and the expressions below integrate over its possible realizations. For a challenger with filing cost K_C , define its equilibrium filing propensity as the probability, over the search outcomes induced by s_C^* , that the realized filing threshold exceeds its challenge cost:

$$\pi_f(K_C; b_E, m_2, R_2) \equiv \Pr_{X_C | s_C^*(K_C)} \left(K_C \leq \hat{K}_C(b_E; m_3, R_3) \middle| m_2, R_2 \right).$$

The applicant's ex ante probability of filing is therefore

$$p_f(b_E; m_2, R_2) \equiv \int \pi_f(K_C; b_E, m_2, R_2) dG_C(K_C).$$

For the same challenger, define the probability of both filing and successfully challenging the patent as

$$\pi_\ell(K_C; b_E, m_2, R_2) \equiv \pi_f(K_C; b_E, m_2, R_2) \mathbb{E}_{X_C | s_C^*(K_C)} \left[\bar{p}_C(b_E; m_3, R_3) \middle| K_C \leq \hat{K}_C(b_E; m_3, R_3), m_2, R_2 \right],$$

with $\pi_\ell(K_C; b_E, m_2, R_2) = 0$ whenever $\pi_f(K_C; b_E, m_2, R_2) = 0$. The applicant's ex ante probability of both facing and losing a challenge is then

$$p_\ell(b_E; m_2, R_2) \equiv \int \pi_\ell(K_C; b_E, m_2, R_2) dG_C(K_C).$$

Thus, both probabilities integrate over the applicant's uncertainty about the challenger's privately observed filing cost and the search outcomes induced by the corresponding equilibrium search choice.³ The probability p_f measures the likelihood of facing a challenge, while p_ℓ measures the likelihood of both facing and losing that challenge. The applicant's expected payoff from accepting the patent is therefore

$$(1 - p_\ell)V_A(b_E, q) - p_f K_A.$$

The applicant accepts if and only if this expected payoff is non-negative. For $p_f > 0$, rearranging yields the threshold rule

$$K_A \leq \hat{K}_A \equiv \frac{1 - p_\ell}{p_f} V_A(b_E, q).$$

³Appendix A provides a more detailed discussion of the construction and interpretation of p_f and p_ℓ .

Thus, the applicant accepts when the value of the granted patent, adjusted for expected challenge and loss risk, is sufficiently large relative to defense cost. If $p_f = 0$, then $p_\ell = 0$, and the applicant accepts whenever $V_A \geq 0$.

5.3 The Examiner Stage

The examiner moves after applicant search and before the applicant's accept-or-abandon decision. After observing the applicant-generated record (m_1, R_1) , the examiner chooses additional search. Search generates s_E additional constraining draws X_E , which are combined with the inherited record to produce

$$m_2(s_E) = \min(X_1 \cup X_E), \quad R_2(s_E) = |X_1 \cup X_E|$$

Under the truncation rule, granted breadth is the most constraining information in the combined record:

$$b_E(s_E) = m_2(s_E).$$

5.3.1 Examiner Search

Given the inherited state (m_1, R_1) , the examiner chooses search intensity

$$s_E \in S = \{0, 1, \dots, \bar{s}\},$$

with cost $c_E(s_E; \lambda_E)$, where search costs are increasing and convex in effort. The examiner's objective is to minimize expected overbreadth loss net of search costs:

$$U_E(s_E; \theta_1) = -\mathbb{E}[V_E(b_E(s_E) - b^*) \mid m_1, R_1] - c_E(s_E; \lambda_E),$$

where

$$\theta_1 = (m_1, R_1, \lambda_E)$$

denotes the examiner's inherited state. Thus, examiner search is valuable only insofar as the induced post-search record reduces expected overbreadth enough to justify the additional search cost.

5.3.2 Existence of an Examiner Search Best Response

The examiner's search problem admits a solution because, under the maintained assumptions, $U_E(s_E; \theta_1)$ is well defined and real valued for every $s_E \in S$, and S is finite. Therefore,

$$BR_E(\theta_1) \equiv \arg \max_{s_E \in S} U_E(s_E; \theta_1)$$

is nonempty. Because the best-response correspondence may contain multiple elements, I define its lower and upper selections as

$$s_E^L(\theta_1) \equiv \min BR_E(\theta_1) \quad \text{and} \quad s_E^H(\theta_1) \equiv \max BR_E(\theta_1).$$

When I refer to equilibrium examiner search below, s_E^* denotes one such selection.

5.4 The Applicant Stage

The applicant moves first strategically. The applicant begins from the initial information state (m_0, R_0) and chooses search intensity s_A . Recall, there is no separate disclosure or breadth-choice stage.

5.4.1 Applicant Search

The applicant chooses

$$s_A \in S = \{0, 1, \dots, \bar{s}\},$$

with cost $c_A(s_A; \lambda_A)$, where search costs are increasing and convex in effort. Search generates additional constraining draws X_A , which are combined with the applicant's initial information to form

$$m_1(s_A) = \min(X_0 \cup X_A), \quad R_1(s_A) = |X_0 \cup X_A|.$$

Applicant search is a value-of-information problem over the continuation game beginning from the state (m_1, R_1) . Let

$$\theta_0 = (q, \lambda_A, K_A, m_0, R_0).$$

Because the applicant can abandon after examination, the continuation value from the applicant-generated state (m_1, R_1) is

$$Z(m_1, R_1; \theta_0) = \mathbb{E} \left[\max \left\{ 0, V_A(b_E, q) \left[1 - p_f(b_E, q; m_2, R_2) (1 - \bar{p}_A^M(b_E; m_2, R_2)) \right] - p_f(b_E, q; m_2, R_2) K_A \right\} \mid m_1, R_1 \right],$$

where the expectation is over examiner and challenger continuation play induced by (m_1, R_1) .

The applicant's ex ante search problem is therefore to choose s_A to maximize

$$U_A(s_A; \theta_0) = \mathbb{E} \left[Z(m_1(s_A), R_1(s_A); \theta_0) \mid m_0, R_0 \right] - c_A(s_A; \lambda_A),$$

where the expectation is over the random outcome of applicant search. Thus, applicant search is valuable because it changes the continuation value of the prosecution and challenge game. Unlike examiner search, which directly reduces expected overbreadth, or challenger search, which changes the expected value of filing, applicant search shapes the information state inherited by all later players.

5.4.2 Existence of an Applicant Search Best Response

The applicant's search problem admits a solution because, under the maintained assumptions, $U_A(s_A; \theta_0)$ is well defined and real valued for every $s_A \in S$, and S is finite. Therefore,

$$BR_A(\theta_0) \equiv \arg \max_{s_A \in S} U_A(s_A; \theta_0)$$

is nonempty. Because the best-response correspondence may contain multiple elements, I define its lower and upper selections as

$$s_A^L(\theta_0) \equiv \min BR_A(\theta_0) \quad \text{and} \quad s_A^H(\theta_0) \equiv \max BR_A(\theta_0).$$

When I refer to equilibrium applicant search below, s_A^* denotes one such selection.

5.5 Equilibrium

The stagewise characterizations above can be combined to construct a subgame-perfect Nash equilibrium.

Proposition 1 *Suppose the search-action set S is finite and the search technology, cost functions, and posterior system $F(\cdot \mid m, R)$ are such that continuation payoffs are well-defined at every admissible history. There exists an equilibrium in which applicant, examiner, and challenger search are chosen from their respective best-response correspondences, granted breadth satisfies $b_E = m_2$, applicant acceptance and challenger filing are governed by threshold rules, and beliefs about the true legally valid breadth are given by $F(\cdot \mid m, R)$ for every information record.*

The existence argument follows directly from backward induction. Applicant acceptance and challenger filing have already been characterized by threshold rules, while the finiteness of S ensures each player has at least one optimal search action whenever continuation payoffs are well-defined. The resulting equilibrium strategy profile is

$$(\{s_A^*, Y_A^*\}, \{s_E^*\}, \{s_C^*, Y_C^*\}),$$

where s_A^* , s_E^* , and s_C^* are the players' search rules, Y_A^* is the applicant's accept-or-abandon rule, and Y_C^* is the challenger's filing rule.

I briefly recap the equilibrium, starting with the challenger. At every inherited post-grant state, the challenger chooses its optimal search intensity. Conditional on the resulting post-search record (m_3, R_3) , the challenger files according to the threshold rule $K_C \leq \hat{K}_C$. Thus, s_C^* and Y_C^* specify optimal choices at every admissible history.

Taking the challenger's continuation behavior as given, the applicant's post-examination accept-or-abandon decision is governed by the threshold rule $K_A \leq \hat{K}_A$. The examiner anticipates this decision and the subsequent challenger stage to choose its optimal search intensity, generating the record (m_2, R_2) . Thus, Y_A^* and s_E^* specify optimal choices at every admissible history.

Finally, the applicant chooses its optimal search intensity, taking as given the examiner's search, its own later acceptance decision, and the challenger's continuation behavior. Thus, s_A^* specifies an optimal choice for every admissible history.

Together, the sequentially optimal choices $(s_A^*, s_E^*, Y_A^*, s_C^*, Y_C^*)$ form an equilibrium. At every on-

or off-path history, beliefs about b^* are determined by the observed record and given by $F(\cdot \mid m, R)$, where (m, R) are the record's sufficient statistics. At every point in the game, each player's choice is optimal, conditional on all later choices being optimal.

6 Comparative Statics

This section studies how equilibrium strategies and outcomes vary with the model's parameters. Because search is discrete and player objectives generally contain threshold or option-value terms, the relevant objective functions need not be differentiable. I therefore use monotone comparative statics rather than first-order conditions. For an ordered search set, an objective $U_i(s_i; z)$ has increasing differences in (s_i, z) if, for any $s'_i > s_i$,

$$U_i(s'_i; z) - U_i(s_i; z)$$

is weakly increasing in z . In that case, a higher z makes higher search relatively more attractive. Decreasing differences are defined analogously and imply that a higher z makes higher search relatively less attractive. With the finite ordered search set used here, these properties allow the extremal best-response selections to be ordered using Topkis's Theorem [37].

I first derive stagewise comparative statics by varying one argument of a player's decision problem, holding the remaining arguments fixed. Next, I assess global effects that propagate through the sequence of play.

6.1 Stagewise Comparative Statics of the Challenger's Filing Decision

Conditional on the realized post-search record, the challenger files according to the threshold rule

$$K_C \leq \hat{K}_C(b_E; m_3, R_3) = V_C(b_E, q) \bar{p}_C(b_E; m_3, R_3).$$

Holding R_3 fixed, a higher post-search minimum m_3 shifts beliefs toward greater legally valid breadth and lowers expected overbreadth. The expected probability of successful challenge therefore falls, reducing $\hat{K}_C$. Equivalently, a more constraining realization of challenger search, represented by a lower m_3 , raises the filing threshold.

Holding m_3 fixed, a larger post-search record R_3 strengthens the screening effect. Under the

maintained record-size ordering, the additional information shifts beliefs toward greater legally valid breadth and lowers expected challenge success. The filing threshold therefore falls as R_3 increases.

Holding the post-search record fixed, higher patent quality q raises $V_C(b_E, q)$ and therefore raises the filing threshold. A higher granted breadth b_E likewise raises the threshold by increasing the value of a successful challenge and, holding beliefs fixed, increasing overbreadth $b_E - b^*$ and the expected probability of challenge success.

Combining these effects, the equilibrium filing threshold $\hat{K}_C$ is weakly increasing in q and b_E , and weakly decreasing in m_3 and R_3 . The challenger's equilibrium filing rule therefore expands the set of challenge costs for which filing occurs when q or b_E rises and contracts it when m_3 or R_3 rises.

6.2 Stagewise Comparative Statics of the Challenger's Search Decision

The challenger's search objective can be written as

$$U_C(s_C; \theta_2) = B_C(s_C; \theta_2) - c_C(s_C; \lambda_C),$$

where the gross expected value of challenger search is

$$B_C(s_C; \theta_2) \equiv \mathbb{E}[\max\{0, V_C(b_E, q)\bar{p}_C(b_E; m_3, R_3) - K_C\} | m_2, R_2].$$

For any $s'_C > s_C$, the incremental gross value of additional search is

$$\Delta B_C(s'_C, s_C; \theta_2) \equiv B_C(s'_C; \theta_2) - B_C(s_C; \theta_2).$$

This difference plays the role of the marginal return to search in the discrete search problem. The search-cost shifter λ_C enters only through $c_C(s_C; \lambda_C)$. By the maintained increasing-differences property of the cost function, a higher λ_C raises the incremental cost of moving from s_C to s'_C . The challenger objective therefore exhibits decreasing differences in (s_C, λ_C) .

The model's primitives provide the economic channels underlying the remaining challenger-search comparative statics. Higher patent quality raises the value of a successful challenge, a lower realized challenge cost expands the set of post-search states in which filing is profitable, a larger inherited record strengthens the screening effect, and a higher inherited minimum $b_E \equiv m_2$ raises both patent value and the scope for overbreadth. These channels suggest that challenger search should be

increasing in q and m_2 , and decreasing in K_C and R_2 . Because additional search can operate through either the boundary effect or the screening effect, however, these level effects do not by themselves determine how the incremental return to higher search changes with each parameter. The following assumption imposes the corresponding ordering on the incremental gross value of search.

Assumption 4 (Challenger Incremental-Search Ordering) *For any $s'_C > s_C$, the incremental gross value $\Delta B_C(s'_C, s_C; \theta_2)$ is weakly increasing in patent quality q and the inherited minimum $m_2 \equiv b_E$, and weakly decreasing in the realized challenge cost K_C and inherited record size R_2 .*

Assumption 4 formalizes these economic channels as restrictions on the incremental expected value of search. It does not require additional search to generate a particular type of information realization; search may still lower the minimum, expand the record without lowering the minimum, or do both.

Under Assumption 4, the challenger objective exhibits increasing differences in (s_C, q) and (s_C, m_2) , and decreasing differences in (s_C, K_C) and (s_C, R_2) . Together with the maintained search-cost condition, it exhibits decreasing differences in (s_C, λ_C) . Topkis's Theorem therefore implies that, taking either extremal best-response selection, equilibrium challenger search $s_C^*(\theta_2)$ is weakly increasing in q and $m_2 \equiv b_E$, and weakly decreasing in λ_C , K_C , and R_2 . Table 1 summarizes these stagewise comparative statics.

6.3 Stagewise Comparative Statics of Applicant Acceptance

The applicant accepts the granted patent whenever

$$K_A \leq \hat{K}_A \equiv \frac{1 - p_\ell}{p_f} V_A(b_E, q),$$

where p_f is the ex ante probability that a challenge is filed and p_ℓ is the ex ante probability that a challenge is both filed and successful. For $p_f > 0$, a reduction in either source of downstream risk raises the accept threshold when the other is held fixed. More generally,

$$p'_f \leq p_f, \quad p'_\ell \leq p_\ell \quad \implies \quad \frac{1 - p'_\ell}{p'_f} \geq \frac{1 - p_\ell}{p_f},$$

holding applicant value fixed. Thus, changes that weakly reduce both filing risk and filing-and-loss risk make acceptance weakly more attractive.

Table 1: Stagewise comparative statics for the challenger

Parameter	Equilibrium search s_C^*	Conditional filing threshold $\hat{K}_C$	Main channel
λ_C	$\downarrow$	$-$	A higher search-cost shifter raises the incremental cost of challenger search.
q	$\uparrow$	$\uparrow$	Higher patent quality raises the value of successful challenge and the incremental value of search.
R_2	$\downarrow$	$-$	A larger inherited record reduces the incremental value of further search through the screening channel.
$m_2 \equiv b_E$	$\uparrow$	$\uparrow$	Greater granted breadth raises challenge value and expected challenge success and increases the incremental value of search.
K_C	$\downarrow$	$-$	A higher realized challenge cost reduces the filing option value and the incremental value of acquiring information.
m_3	$-$	$\downarrow$	A higher post-search minimum reduces expected overbreadth and expected challenge success.
R_3	$-$	$\downarrow$	A larger post-search record strengthens screening and lowers expected challenge success holding the minimum fixed.

Notes: Arrows indicate the effect of an increase in the parameter listed in the first column. Search effects for q , R_2 , m_2 , and K_C use Assumption 4; the effect of λ_C follows from the maintained search-cost assumption. The filing threshold is

$$\hat{K}_C(b_E; m_3, R_3) = V_C(b_E, q) \bar{p}_C(b_E; m_3, R_3).$$

Threshold effects hold the realized post-search record fixed and therefore exclude effects operating through equilibrium challenger search.

Recall that the ex ante filing probability is

$$p_f(b_E; m_2, R_2) = \int \pi_f(K_C; b_E, m_2, R_2) dG_C(K_C),$$

where $\pi_f(K_C; b_E, m_2, R_2)$ is the equilibrium filing propensity of a challenger with realized cost K_C . Higher-cost challenger types are weakly less likely to file:

$$K'_C > K_C \implies \pi_f(K'_C; b_E, m_2, R_2) \leq \pi_f(K_C; b_E, m_2, R_2).$$

Intuitively, a higher filing cost reduces the set of challenger-search realizations in which filing is profitable. This ordering continues to hold when the challenger adjusts its equilibrium search choice because the challenger's optimized continuation value is convex in K_C .⁴

The same reasoning does not necessarily order the probability of both filing and winning. A higher-cost challenger may file less often while selecting into challenges with a greater conditional probability of success. The following assumption limits the strength of this selection effect.

Assumption 5 (Challenge-Cost Dominance) *For any $K'_C > K_C$, the type-specific filing-and-loss probability satisfies $\pi_\ell(K'_C; b_E, m_2, R_2) \leq \pi_\ell(K_C; b_E, m_2, R_2)$.*

Assumption 5 requires higher challenge costs to weakly reduce the overall probability that the challenger both files and successfully challenges the patent. It does not require the conditional success probability of filed challenges to fall with K_C . Higher-cost challengers may select into stronger cases, provided that this selection effect does not outweigh the reduction in filing.

Consider now a first-order stochastic shift in the challenge-cost environment toward higher costs. Because both $\pi_f(K_C)$ and, under Assumption 5, $\pi_\ell(K_C)$ are weakly decreasing in K_C , such a shift weakly lowers both p_f and p_ℓ . Both changes raise $\hat{K}_A$, so a higher challenge-cost environment makes applicant acceptance weakly more likely.

The inherited prosecution record R_2 also affects downstream risk. Holding granted breadth fixed, a larger inherited record strengthens the screening effect and directly reduces expected overbreadth. At the same time, Assumption 4 implies that a larger R_2 reduces challenger search, and the resulting change in the post-search record need not reinforce the direct screening effect. The following assumption orders the combined effect.

⁴Appendix B provides the formal argument.

Assumption 6 (Inherited-Record Dominance) *For any $R'_2 > R_2$, holding $b_E \equiv m_2$ and q fixed, $\pi_f(K_C; b_E, m_2, R'_2) \leq \pi_f(K_C; b_E, m_2, R_2)$, and $\pi_\ell(K_C; b_E, m_2, R'_2) \leq \pi_\ell(K_C; b_E, m_2, R_2)$ for every K_C .*

Assumption 6 requires the direct screening effect of a larger inherited record to dominate any offsetting effect generated by the induced reduction in challenger search. Integrating over G_C , a larger R_2 therefore weakly lowers both p_f and p_ℓ , raising the applicant's accept threshold.

Patent quality and granted breadth affect acceptance both through applicant value and through downstream challenger behavior. Rather than separately restricting each component of challenge risk, I impose a dominance condition directly on the applicant's risk-adjusted patent value.

Assumption 7 (Applicant Value Dominance) *The applicant's risk-adjusted patent value satisfies the following conditions.*

(i) *For any $q' > q$, holding (b_E, m_2, R_2) fixed,*

$$V_A(b_E, q') \frac{1 - p'_\ell}{p'_f} \geq V_A(b_E, q) \frac{1 - p_\ell}{p_f},$$

where p'_f and p'_ℓ denote the equilibrium challenge probabilities induced by q' .

(ii) *For any $m'_2 > m_2$, with $b'_E = m'_2$ and holding q and R_2 fixed,*

$$V_A(b'_E, q) \frac{1 - p'_\ell}{p'_f} \geq V_A(b_E, q) \frac{1 - p_\ell}{p_f},$$

where p'_f and p'_ℓ denote the equilibrium challenge probabilities induced by $m'_2 \equiv b'_E$.

The first part of Assumption 7 requires the direct increase in applicant value from higher patent quality to dominate any adverse change in downstream challenge risk. The second imposes the analogous restriction for greater granted breadth. Under these conditions, $\hat{K}_A$ is weakly increasing in both q and $m_2 \equiv b_E$.

The effect of the challenger search-cost shifter λ_C is generally ambiguous. A higher λ_C lowers equilibrium challenger search, but less search can weaken either the boundary effect or the screening effect depending on the induced distribution of post-search records. Consequently, the model does not generally order the resulting changes in p_f or p_ℓ , and therefore does not sign the effect of λ_C on applicant acceptance without an additional restriction.

Finally, the applicant's realized defense cost K_A does not shift $\hat{K}_A$. It instead enters directly into the acceptance rule. For $K'_A > K_A$, any patent accepted at K'_A would also be accepted at K_A , so higher defense cost weakly reduces applicant acceptance.

Combining these results, the equilibrium accept threshold $\hat{K}_A$ is weakly increasing following a shift toward higher challenge costs and an increase in R_2 , and, under Assumption 7, is weakly increasing in q and $b_E \equiv m_2$. Its response to λ_C remains ambiguous. Because the applicant accepts whenever $K_A \leq \hat{K}_A$, the equilibrium acceptance decision inherits these monotonicities, while a higher realized defense cost K_A weakly reduces acceptance. Table 2 summarizes the stagewise comparative statics.

Table 2: Stagewise comparative statics for the applicant accept decision

Parameter	p_f	p_ℓ	Accept threshold $\hat{K}_A$	Acceptance	Main channel
λ_C	?	?	?	?	Higher challenger search cost reduces search, but the induced effect on downstream challenge risk is ambiguous.
Higher challenge-cost environment	↓	↓	↑	↑	Higher challenger costs reduce filing; under challenge-cost dominance, they also reduce the probability of both filing and winning.
q	?	?	↑	↑	Higher quality raises applicant value; under applicant value dominance, this effect dominates any adverse change in challenge risk.
$m_2 \equiv b_E$	?	?	↑	↑	Greater granted breadth raises applicant value; under applicant value dominance, this effect dominates any adverse change in challenge risk.
R_2	↓	↓	↑	↑	Under inherited-record dominance, a larger prosecution record reduces both filing and filing-and-loss risk.
K_A	—	—	—	↓	Higher realized defense cost makes acceptance less attractive directly.

Notes: The applicant accepts when

$$K_A \leq \hat{K}_A \equiv \frac{1 - p_\ell}{p_f} V_A(b_E, q).$$

The challenge-cost result for p_f follows from the declining type-specific filing propensity $\pi_f(K_C)$, while the result for p_ℓ uses Assumption 5. The R_2 signs use Assumption 6, and the acceptance effects of q and $m_2 \equiv b_E$ use Assumption 7. Question marks indicate effects that are not generally signed by the maintained assumptions.

6.4 Stagewise Comparative Statics of Examiner Search

I next characterize how the examiner's search choice responds to changes in the inherited record and the examiner's search cost. Write the examiner's objective as

$$U_E(s_E; \theta_1) = B_E(s_E; \theta_1) - c_E(s_E; \lambda_E),$$

where $B_E(s_E; \theta_1)$ is the gross expected value of examiner search. For any $s'_E > s_E$, define

$$\Delta B_E(s'_E, s_E; \theta_1) \equiv B_E(s'_E; \theta_1) - B_E(s_E; \theta_1).$$

The search-cost shifter λ_E enters only through $c_E(s_E; \lambda_E)$. By the maintained increasing-differences property of the cost function, a higher λ_E raises the incremental cost of moving from s_E to s'_E . The examiner objective therefore exhibits decreasing differences in (s_E, λ_E) .

The model's information structure provides the economic channels underlying the remaining examiner-search comparative statics. Holding m_1 fixed, a larger inherited record R_1 strengthens the screening effect, shifting beliefs toward greater legally valid breadth and lowering expected overbreadth. A higher inherited minimum m_1 , by contrast, means that broader proposed breadth enters examination. Under the maintained minimum-dominance condition, this increase in breadth dominates the corresponding increase in expected true breadth, raising expected overbreadth. These channels suggest that examiner search should be decreasing in R_1 and increasing in m_1 . Because the effect of the inherited state on expected overbreadth does not by itself order the incremental return to higher search, I impose the corresponding ordering directly.

Assumption 8 (Examiner Incremental-Search Ordering) *For any $s'_E > s_E$, the incremental gross value $\Delta B_E(s'_E, s_E; \theta_1)$ is weakly increasing in the inherited minimum m_1 and weakly decreasing in the inherited record size R_1 .*

Assumption 8 formalizes these channels as restrictions on the incremental value of examiner search without requiring a particular information realization.

Under Assumption 8, the examiner objective exhibits increasing differences in (s_E, m_1) and decreasing differences in (s_E, R_1) . Together with the maintained search-cost condition, it exhibits decreasing differences in (s_E, λ_E) . Topkis's Theorem therefore implies that, taking either extremal

best-response selection, equilibrium examiner search $s_E^*(\theta_1)$ is weakly increasing in m_1 and weakly decreasing in λ_E and R_1 . Table 3 summarizes these stagewise comparative statics.

Table 3: Stagewise comparative statics for the examiner

Parameter	Equilibrium search s_E^*	Main channel
λ_E	$\downarrow$	A higher examiner search-cost shifter raises the incremental cost of search.
R_1	$\downarrow$	A larger inherited record strengthens screening and, under Assumption 8, lowers the incremental value of additional search.
m_1	$\uparrow$	A higher inherited minimum raises expected overbreadth under minimum dominance and, under Assumption 8, raises the incremental value of additional search.

Notes: Arrows indicate the effect of an increase in the parameter listed in the first column. The effect of λ_E follows from the maintained search-cost assumption. The R_1 and m_1 signs use Assumption 8; the economic motivation for the m_1 ordering additionally relies on the maintained minimum-dominance condition.

6.5 Stagewise Comparative Statics of Applicant Search

I next characterize how applicant search responds to changes in the initial record, applicant incentives, and the incentives facing later players. Because applicant search occurs before examination and challenge, these effects operate through the continuation value $Z(m_1, R_1; \theta_0)$. Write the applicant's objective as

$$U_A(s_A; \theta_0) = B_A(s_A; \theta_0) - c_A(s_A; \lambda_A),$$

where

$$B_A(s_A; \theta_0) \equiv \mathbb{E}[Z(m_1, R_1; \theta_0) | m_0, R_0, s_A].$$

For any $s'_A > s_A$, define

$$\Delta B_A(s'_A, s_A; \theta_0) \equiv B_A(s'_A; \theta_0) - B_A(s_A; \theta_0).$$

The applicant's search-cost shifter λ_A enters only through $c_A(s_A; \lambda_A)$. By the maintained increasing-differences property of the cost function, a higher λ_A raises the incremental cost of moving from s_A

to s'_A . The applicant objective therefore exhibits decreasing differences in (s_A, λ_A) .

The model's primitives provide the economic channels underlying the remaining applicant-search comparative statics. Holding m_0 fixed, a larger initial record R_0 leaves less uncertainty for applicant search to resolve. A higher initial minimum m_0 leaves a less constraining initial record and increases the value of learning whether broader protection is sustainable. Higher patent quality q raises the value of the eventual patent and therefore the stakes associated with the information state carried into examination and challenge. A higher defense cost K_A lowers the continuation value of patents exposed to downstream challenge. Finally, later-player incentives affect the value of shaping the information state before those players act. Higher examiner search costs reduce expected examination-stage scrutiny, while a shift toward higher challenger filing costs weakens downstream challenge pressure. These channels suggest that applicant search should increase in q and m_0 , and decrease in K_A , R_0 , λ_E , and the challenge-cost environment. As before, these level effects do not by themselves order the incremental continuation value generated by higher applicant search. I therefore impose the corresponding ordering directly.

Assumption 9 (Applicant Incremental-Search Ordering) *For any $s'_A > s_A$, the incremental gross value $\Delta B_A(s'_A, s_A; \theta_0)$ is weakly increasing in patent quality q and the initial minimum m_0 , and weakly decreasing in the applicant's defense cost K_A , the initial record size R_0 , and the examiner search-cost shifter λ_E . In addition, a first-order stochastic shift of G_C toward higher challenger filing costs weakly reduces $\Delta B_A(s'_A, s_A; \theta_0)$.*

Assumption 9 formalizes these channels as restrictions on the incremental continuation value of applicant search. In particular, the challenge-cost condition captures the idea that stronger downstream challenge incentives increase the value of acquiring information earlier in the patenting process.

Under Assumption 9, the applicant objective exhibits increasing differences in (s_A, q) and (s_A, m_0) , and decreasing differences in (s_A, K_A) , (s_A, R_0) , and (s_A, λ_E) . Together with the maintained search-cost condition, it exhibits decreasing differences in (s_A, λ_A) . The same ordering implies that a first-order stochastic shift toward higher challenger filing costs lowers the return to greater applicant search. Thus, taking either extremal best-response selection, equilibrium applicant search $s_A^*(\theta_0)$ is weakly increasing in q and m_0 , weakly decreasing in λ_A , K_A , R_0 , and λ_E , and weakly decreasing following a shift toward higher challenger filing costs.

The effect of the challenger search-cost shifter λ_C remains ambiguous. A higher λ_C lowers equilibrium challenger search, but the resulting change in the distribution of post-search records need not systematically raise or lower p_f or p_ℓ . The model therefore does not generally order the resulting change in the applicant's incremental continuation value or in s_A^* . Table 4 summarizes these stagewise comparative statics.

Table 4: Stagewise comparative statics for applicant search

Parameter	Equilibrium search s_A^*	Main channel
λ_A	↓	A higher applicant search-cost shifter raises the incremental cost of search.
q	↑	Higher patent quality raises the stakes associated with the information state and, under Assumption 9, the incremental value of search.
K_A	↓	Higher defense cost lowers the continuation value of patents exposed to challenge and, under Assumption 9, the incremental value of search.
λ_E	↓	Higher examiner search costs weaken expected examination-stage scrutiny and, under Assumption 9, reduce the incremental value of applicant search.
λ_C	?	Higher challenger search costs reduce challenger search, but the resulting effect on downstream challenge risk and the value of applicant search is ambiguous.
Higher challenge-cost environment	↓	Higher challenger filing costs weaken downstream challenge pressure and, under Assumption 9, reduce the incremental value of applicant search.
m_0	↑	A higher initial minimum leaves broader protection potentially sustainable and raises the incremental value of search under Assumption 9.
R_0	↓	A larger initial record leaves less uncertainty to resolve and lowers the incremental value of search under Assumption 9.

Notes: Arrows indicate the effect of an increase in the parameter listed in the first column. The λ_A sign follows directly from the maintained search-cost assumption. The signs for q , K_A , λ_E , m_0 , R_0 , and the challenge-cost environment use Assumption 9. A higher challenge-cost environment refers to a first-order stochastic shift of G_C toward higher filing costs. The effect of λ_C is not generally signed by the maintained assumptions.

6.6 Global Monotone Comparative Statics

The stagewise comparative statics above characterize the direct responses within each player's decision problem. Global comparative statics are harder because a parameter change that moves

applicant search also changes the record inherited by the examiner, which then induces an endogenous examiner response. To obtain sharp signs for the on-path prosecution record, I impose one additional restriction on how strongly examiner search offsets changes in applicant search.

The restriction is consistent with the institutional fact that patent examination is shaped by binding time allocations and workload pressure [12, 11, 18]. In the model, the same idea follows from diminishing returns to examiner search and increasing marginal search costs: examiner search may adjust in response to applicant search, but it does not fully undo it.

Assumption 10 (Partial Examiner Adjustment) *For any parameter change that alters equilibrium applicant search, the induced change in equilibrium examiner search is weakly in the opposite direction and strictly smaller in absolute value whenever applicant search changes. Formally, if a parameter change from τ to τ' changes equilibrium search from $s_A^*(\tau)$ to $s_A^*(\tau')$ and from $s_E^*(\tau)$ to $s_E^*(\tau')$, then $(s_A^*(\tau') - s_A^*(\tau))(s_E^*(\tau') - s_E^*(\tau)) \leq 0$ and, whenever $s_A^*(\tau') \neq s_A^*(\tau)$, $|s_E^*(\tau') - s_E^*(\tau)| < |s_A^*(\tau') - s_A^*(\tau)|$.*

Let $\Delta x \equiv x(\tau') - x(\tau)$ denote the equilibrium change induced by a parameter shift. Assumption 10 implies total pre-grant search moves in the same direction as applicant search:

$$\Delta s_A^* > 0 \implies \Delta(s_A^* + s_E^*) > 0,$$

and

$$\Delta s_A^* < 0 \implies \Delta(s_A^* + s_E^*) < 0.$$

Because the prosecution record is built from cumulative pre-grant search, the inherited record size R_2 moves in the same direction as s_A^* . By the maintained information structure, higher cumulative search also shifts the effective minimum m_2 in a more constraining direction, while lower cumulative search shifts it in a less constraining direction. The examiner therefore smooths the effect of applicant search on the prosecution record but does not overturn it.

The applicant's accept-or-abandon decision requires one further dominance restriction. A stronger prosecution record tends to reduce downstream filing and filing-and-loss risk, but it may also reduce granted breadth and therefore lower patent value. Moreover, some parameter changes may affect downstream challenge incentives directly in addition to operating through the prosecution record.

I maintain that, after accounting for these direct and indirect equilibrium effects, the change in the applicant's risk-adjusted continuation value is dominated by the change in the strength of the prosecution record. Thus,

$$\Delta s_A^* > 0 \implies \Delta \left[V_A(b_E, q) \frac{1 - p_\ell}{p_f} \right] \geq 0,$$

while

$$\Delta s_A^* < 0 \implies \Delta \left[V_A(b_E, q) \frac{1 - p_\ell}{p_f} \right] \leq 0.$$

Since this risk-adjusted value is the applicant's accept threshold $\hat{K}_A$, applicant acceptance inherits the direction of the prosecution record under the maintained continuation-value dominance restriction.

The challenger-side objects are less sharp. A parameter change can affect challenger incentives directly while also changing applicant search and thereby altering the prosecution record inherited by the challenger. These channels jointly determine equilibrium challenger search and the type-specific filing and loss propensities, and therefore the ex ante probabilities p_f and p_ℓ . The direct challenger-side effect and the indirect effect through the prosecution record may point in opposite directions. Without an additional dominance restriction on their relative magnitudes, the global effects on challenger search, filing probability, and filing-and-loss probability remain ambiguous.

These observations can be summarized as follows.

Proposition 2 *Under Assumption 10 and the maintained continuation-value dominance restriction, any parameter change that raises equilibrium applicant search raises cumulative pre-grant search, increases the prosecution record size, shifts the effective minimum in a more constraining direction, and weakly raises applicant acceptance. Any parameter change that lowers equilibrium applicant search has the opposite effects on the prosecution record and weakly lowers applicant acceptance. The global effects on challenger search, filing probability, and filing-and-loss probability remain ambiguous in general.*

Proposition 2 gives the basic structure of the global comparative statics. The sharpest global predictions concern the prosecution-stage information state and the applicant-side equilibrium objects. Challenger-side outcomes remain more sensitive to offsetting equilibrium feedback because

applicant behavior itself acts as an endogenous deterrence margin.

Table 5 collects the global comparative statics for the main equilibrium objects. These signs describe how changes in primitives propagate through applicant search, examiner scrutiny, granted breadth, acceptance, and post-grant challenge behavior.

Table 5: Global monotone comparative statics

Parameter	s_A^*	m_1	s_E^*	m_2	Acceptance	s_C^*	pf
q	$\uparrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\uparrow$	$?$	$?$
λ_A	$\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	$?$	$?$
λ_E	$\downarrow$	$\uparrow$	$?$	$\uparrow$	$\downarrow$	$?$	$?$
λ_C	$?$	$?$	$?$	$?$	$?$	$?$	$?$
FOSD $\uparrow$ in G_C	$\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	$?$	$?$
FOSD $\uparrow$ in G_A	$\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	$?$	$?$
m_0	$\uparrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\uparrow$	$?$	$?$
R_0	$\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	$?$	$?$

Notes: The signs summarize the global equilibrium effects in Proposition 2. A decrease in m_1 or m_2 means the relevant minimum shifts in a more constraining direction. Under Assumption 10, changes in applicant search move the prosecution-stage record in the same direction as cumulative pre-grant search. The sign on acceptance uses the maintained continuation-value dominance restriction. Question marks indicate cases in which direct incentive effects and indirect equilibrium-record effects may offset one another. In the λ_E row, examiner search is ambiguous because higher examiner search costs lower s_E^* directly, while the induced reduction in applicant search creates an offsetting examiner response. In the λ_C row, the effect on applicant search is itself ambiguous, so the induced prosecution-stage and acceptance effects cannot generally be signed.

7 Empirical Content and Descriptive Evidence

This section connects the model to observable patent data. The model’s equilibrium objects are not observed directly, but they correspond to standard measures of applicant disclosures, examiner-added information, claim breadth, prosecution outcomes, and post-grant challenges. These mappings generate testable predictions for how changes in search costs, challenge costs, and patent value should appear in the data.

7.1 From Model Objects to Empirical Proxies

The empirical content of the model turns on separating three margins: applicant-generated information, examiner-added information, and downstream challenge activity. Applicant and examiner search are not directly observed, and legally valid breadth b^* is latent. The goal is therefore not to find one perfect proxy for each model object, but to organize observable patent-data measures

according to the sequence of the game.

Applicant search s_A is best interpreted as applicant-side information production. Its closest empirical counterparts are applicant-submitted prior-art references, IDS filings, applicant-submitted patent citations, and applicant-submitted non-patent literature. These variables measure information placed into the prosecution record rather than latent search effort itself, since U.S. law imposes a duty to disclose known material information but not a general duty to conduct a prior-art search. The empirical counterpart of R_1 is therefore the size of the applicant-generated record observed by the examiner. Prior work emphasizes the importance of separating applicant-submitted and examiner-added references because they capture different parts of the prosecution process [1, 16, 7].

The applicant-stage minimum m_1 corresponds to the breadth of the claims entering examination. Following the claim-breadth literature, this object can be proxied by first independent claim length, shortest independent claim length, or the number of independent claims at application or publication [28, 17]. The maintained interpretation is that, all else equal, shorter claims or more independent claiming routes tend to indicate broader breadth. Because claim-drafting conventions vary across technologies, these proxies should be used with technology fixed effects, art-unit controls, and robustness checks for fields where claim length may be less informative [28, 29].

Examiner search s_E maps into examiner-side information production and prosecution-stage scrutiny. Natural proxies include examiner-added citations, rejection art, office actions, Section 102 and Section 103 rejections, RCEs, appeals, interviews, and prosecution duration. Examiner-added citations are closest to the model’s information-production margin, while office actions and rejections capture broader prosecution intensity. Prior work documents systematic variation across examiners and institutional environments, supporting the interpretation of examination as an endogenous screening margin [5, 22, 12, 11, 10].

Granted breadth $b_E = m_2$ corresponds to claim breadth at grant. In the data, m_2 can be proxied using grant-stage claim-breadth measures such as first independent claim length, shortest independent claim length, or independent claim count. The difference between application-stage and grant-stage breadth provides a measure of prosecution-stage narrowing [28, 4, 29]. This narrowing reflects both examiner objections and applicant amendment choices, so it should be interpreted as an equilibrium prosecution outcome rather than a pure measure of examiner effort.

The applicant accept-or-abandon decision maps into whether the application continues through prosecution and ultimately issues as a patent. Observable outcomes include grant, abandonment,

continuation filings, continuation-in-part filings, divisional applications, RCEs, and appeals. These outcomes are useful because the model predicts that search and challenge risk affect not only granted breadth but also the applicant’s willingness to continue pursuing protection.

The challenger stage is observed through post-grant activity. Filing Y_C and filing probability p_f map into district court litigation, IPR petitions, PGR petitions, institution decisions, final written decisions, and claim cancellation or amendment. Challenger search s_C is harder to observe, but asserted prior art in PTAB petitions or litigation provides a partial measure of challenger-side information production. Post-grant proceedings are especially useful because they reveal whether challengers rely on art already in the prosecution record or instead identify new prior art not previously considered [27, 3, 44].

Table 6 summarizes the main empirical counterparts. The mapping is not meant to imply that any single variable perfectly measures the corresponding theoretical object. Rather, the table organizes standard patent-data measures according to the sequence of the model.

7.2 Descriptive Facts Motivating the Empirical Margins

Three descriptive facts from the existing literature motivate the empirical relevance of treating claim breadth, applicant-generated information, examiner-generated information, and post-grant challenge as distinct but connected margins.

First, downstream challenges are selective. If all granted patents faced the same post-grant threat, the challenger stage would have little empirical content. In practice, challenges are concentrated on patents with high stakes and high expected returns to invalidation. Prior work on IPR and related proceedings shows challenged patents differ systematically from unchallenged patents along measures of breadth, value, and legal vulnerability [27, 3]. Martinelli and Mazzei provide closely related evidence using claim-breadth measures, showing that patents subject to post-grant opposition tend to have broader claims than otherwise similar non-opposed patents [29]. This fact corresponds to the filing condition in the model: broader and more valuable patents raise the return to challenge and may increase expected success by increasing expected overbreadth.

Second, post-grant challenge institutions can affect behavior before any challenge is filed. This is the upstream discipline mechanism in the model. A change in the cost or credibility of challenge changes the expected payoff from seeking broad claims and can therefore alter applicant and examiner behavior during prosecution. Martinelli and Mazzei study the introduction of post-grant

Table 6: Model objects and empirical counterparts

Model object	Interpretation	Empirical counterparts
s_A	Applicant search / applicant information production	Applicant-submitted citations; IDS filings; applicant patent citations; applicant non-patent literature citations; references submitted before or during prosecution.
R_1	Size of the applicant-generated record	Number of applicant-submitted references; number of IDS filings; applicant-submitted patent and non-patent references.
m_1	Breadth carried into examination	First independent claim length at application or publication; shortest independent claim length; independent claim count; application-stage claim-breadth measures.
s_E	Examiner search and scrutiny	Examiner-added citations; examiner-used rejection art; office actions; Section 102 and Section 103 rejections; interviews; appeals; RCEs; prosecution duration.
$R_2 - R_1$	Examiner-added record	Examiner-added references; examiner citations used in rejection; additional prior art introduced during prosecution.
$m_2 = b_E$	Granted breadth	First independent claim length at grant; shortest independent claim length at grant; independent claim count at grant; grant-stage claim-breadth measures.
$m_1 - m_2$ or Δm	Prosecution-stage narrowing	Change in claim length from application or publication to grant; amendments to independent claims; reduction in independent claim count; difference between application and grant breadth.
Y_A	Applicant acceptance / continuation	Grant; abandonment; continuation filings; continuation-in-part filings; divisional filings; RCEs; appeals.
s_C	Challenger search	Prior art asserted in IPR, PGR, or litigation; newly introduced references in post-grant proceedings; overlap between asserted art and prosecution-history art.
Y_C and p_f	Challenge filing and challenge probability	IPR petitions; PGR petitions; district court litigation; institution decisions; final written decisions; claim cancellation or amendment.
q	Patent value / stakes of breadth	Forward citations; family size; PCT filing; renewal or maintenance behavior; assignment or transaction measures.
λ_A	Applicant search-cost shifter	Applicant sophistication; attorney characteristics; prior-art complexity; technology field; firm size; experience with patent prosecution.
λ_E	Examiner search-cost shifter	Examiner workload; examiner seniority; art-unit backlog; time allocations; technology-specific prior-art complexity; institutional examination constraints.
λ_C	Challenger search-cost shifter	Litigation or PTAB search costs; availability of prior-art search tools; technology-specific prior-art complexity; party sophistication; challenger resources.
K_C and K_A	Challenge and defense costs	AIA/PTAB availability; forum; technology-level litigation cost; party type; entity size; NPE involvement; expected cost of challenge.

review through the AIA using U.S.-Canada patent twins and find that applicants respond by filing narrower claims at the USPTO after the reform. They also find greater prosecution-stage narrowing when applicants do not preemptively narrow claims themselves, consistent with examiners partly offsetting applicant inaction [29]. In the language of the model, a change in the downstream challenge environment shifts not only Y_C or p_f , but also upstream objects such as m_1 , m_2 , and the prosecution record inherited by later players.

Third, applicant-submitted information and examiner-generated information are empirically distinct. The model treats applicant search and examiner search as separate margins because the applicant-generated record is observed by the examiner and then augmented during examination. Patent-citation evidence supports this separation. Applicant-submitted and examiner-added references differ in timing, source, and empirical interpretation [1]. Prior work also shows that applicant and examiner citations have different relationships with patent value and prosecution outcomes [16, 7]. This distinction matters because total prior art in the record is not enough to identify the mechanism: a richer prosecution record may reflect applicant search, examiner search, or both.

Together, these facts support studying the empirical margins jointly rather than treating claim breadth, prosecution information, and post-grant challenge as unrelated outcomes.

7.3 Testable Predictions

I now use the comparative statics to organize a set of testable predictions for an empirical implementation of the theory.

A natural empirical setting is the introduction of post-grant review through the America Invents Act. In the language of the model, the AIA can be interpreted primarily as a shift toward a lower challenge-cost environment, since it created new administrative procedures for challenging issued patents through the PTAB. This interpretation has two caveats. First, the AIA was not a single-parameter change; it also included other institutional changes, most notably the transition from first-to-invent to first-to-file. Second, the relevant timing may differ across players. Applicants may respond when the legal change becomes anticipated, while examiners may respond more strongly once post-grant review becomes effective and their examination outcomes can be scrutinized through the new procedures [29]. The AIA is therefore best viewed as a useful setting for studying changes in challenge risk, not as a perfect experiment that moves only one primitive of the model.

The predictions should be interpreted as weak directional predictions. When the model predicts

an increase or decrease in an equilibrium object, the empirical implication is that the object should move in the predicted direction or display no meaningful response. Some margins may appear close to zero because the relevant constraint does not bind, the treatment is weak for patents unlikely to face challenge, or offsetting equilibrium channels make the net response small. The empirical content of the model is therefore not that every margin responds strongly in every sample, but that systematic responses should align with the directional logic of the comparative statics.

The first set of predictions concerns applicant behavior. When downstream challenge becomes more severe, applicants have stronger incentives to learn about the limits of valid protection before examination. In the model, this appears as higher applicant search s_A and narrower breadth entering examination, m_1 . Empirically, this predicts weak increases in the applicant-side information measures summarized in Table 6 and weak narrowing of application-stage claims, measured by longer first independent claims, longer shortest independent claims, or fewer independent claiming routes.

The second set of predictions concerns examiner scrutiny and the prosecution record. Because applicant and examiner search interact, a stronger applicant response can reduce the marginal value of examiner search. Under the partial-adjustment condition used for the global comparative statics, lower challenge costs increase applicant search and strengthen the applicant-generated record, while examiner search may fall in response. Nevertheless, total pre-grant information increases and granted breadth falls. Empirically, examiner-side search measures should therefore be interpreted jointly with the combined prosecution record and the amount of narrowing between application and grant. A weak decline in examiner-side search need not imply weaker screening if it occurs alongside a stronger applicant-generated record and narrower granted claims.

The third set of predictions concerns grant outcomes and post-grant challenge. The model predicts that a stronger prosecution record can increase the applicant's willingness to accept the resulting patent because the patent is less vulnerable conditional on challenge. Grant, abandonment, continuation, and RCE outcomes are therefore relevant applicant-side margins. The prediction for observed post-grant challenges is less sharp. A lower challenge-cost environment directly makes filing more attractive to challengers, but it also induces upstream responses that may make granted patents less vulnerable. Thus, PTAB availability need not mechanically increase observed challenge rates. The observed effect on post-grant proceedings depends on the relative strength of the direct challenge-cost effect and the indirect upstream-discipline effect.

Taken together, the model predicts a coordinated response across upstream information pro-

duction, claim breadth, prosecution outcomes, and downstream challenge rather than a response confined to observed post-grant filings.

7.4 Empirical Implementation in Future Work

These predictions can be implemented by combining patent text, prosecution-history data, citation-source information, and post-grant challenge outcomes.

One empirical setting is the introduction of post-grant review through the AIA. A simple design would compare patents more and less exposed to post-grant challenge risk before and after the reform. Exposure could be measured using pre-reform technology-level litigation or challenge rates, patent value, claim breadth, patent-thicket density, or other predictors of downstream challenge. The basic specification could relate an outcome Y_{it} to the interaction between the post-AIA period and predicted challenge exposure:

$$Y_{it} = \beta (\text{PostAIA}_t \times \text{Threat}_i) + X'_{it}\Gamma + \alpha_{j(i)} + \lambda_t + \varepsilon_{it},$$

where Y_{it} may measure applicant-submitted prior art, examiner-added prior art, application claim breadth, prosecution-stage narrowing, grant or abandonment, or post-grant challenge. Technology fixed effects $\alpha_{j(i)}$ and time fixed effects λ_t absorb persistent differences across technological areas and aggregate changes in patenting behavior.

A related design could follow the patent-twins approach used in prior work and compare U.S. applications to foreign counterparts for the same underlying invention [29]. Family fixed effects would absorb invention-level heterogeneity, with identifying variation coming from whether the U.S. member of the family is exposed to the post-AIA challenge environment. A richer version of this design could interact U.S. exposure after the reform with predicted challenge threat:

$$Y_{ift} = \beta (\text{US}_i \times \text{PostAIA}_t \times \text{Threat}_f) + \alpha_f + \lambda_t + \varepsilon_{ift},$$

where α_f denotes patent-family fixed effects. This design is especially useful for studying claim breadth and prosecution-stage narrowing because it compares the same invention across legal environments.

These designs illustrate how the model can discipline both the outcomes examined and the interpretation of responses across different stages of the patenting process.

8 Conclusion

Patent breadth is often discussed as if it were determined either by the applicant’s initial claim-drafting decision or by the patent office’s grant decision. Instead, I argue that patent breadth is the equilibrium outcome of a sequential search process involving applicants, examiners, and potential challengers. Applicants decide how much information to acquire before examination, examiners decide how much additional information to acquire during prosecution, and challengers decide whether further search and filing are worthwhile after grant. The breadth that ultimately matters is therefore shaped by strategic information acquisition across the entire patenting sequence.

The model formalizes this idea in a three-player dynamic game. The central state variable is the prosecution record, summarized by (m, R) , where m is the most constraining draw and R is the size of the accumulated record. This structure captures two roles of search. Search can uncover a more constraining piece of prior art, lowering the breadth that can be supported, and it can enlarge the record, changing how later players form beliefs about the legally valid breadth of the patent. These two channels link applicant search, examiner search, and challenger search through a common information structure.

The main implication is that the stages of the patenting process cannot be understood independently. Applicant search shapes the record inherited by the examiner and later by the challenger. Examiner search affects both the breadth that survives prosecution and the public record on which downstream challenge decisions are based. Challenger search and filing occur after grant, but the expected threat of challenge feeds back into applicant and examiner incentives before grant. As a result, post-grant institutions can alter upstream behavior even before any challenge is filed.

The comparative statics highlight this equilibrium feedback. The sharpest predictions concern the applicant-generated record, the prosecution-stage information state, and the breadth that survives examination. Parameters that raise applicant search tend to strengthen the prosecution record and shift granted breadth in a more constrained direction, provided examiner adjustment is only partial. By contrast, the effect on observed post-grant challenge is generally ambiguous. Lower challenge costs directly make filing more attractive to challengers, but they can also induce upstream responses that make issued patents less vulnerable. Observed filing therefore depends on the balance between the direct challenge-cost effect and the indirect deterrence effect operating through applicant and examiner behavior.

This logic has implications for patent policy. Post-grant review should not be evaluated only by counting challenges filed or claims invalidated after grant. Those outcomes are important, but they are the final part of a longer strategic sequence. A reform that changes the cost or credibility of challenge may also change applicant search, the information placed into the prosecution record, examiner scrutiny, claim narrowing, and the set of applications applicants choose to continue pursuing. Similarly, policies that affect examiner capacity or applicant search incentives can have downstream effects on challenge behavior because they change the record later challengers inherit. The relevant policy environment is therefore the joint system of pre-grant and post-grant incentives.

The model also clarifies why patent breadth is a useful object for studying patent quality. A binary distinction between valid and invalid patents misses much of the economic action. Many patents matter because of how far their claims extend, how much uncertainty they create for rivals, and how costly they are to challenge. By focusing on breadth, the framework connects prior-art search, examination, and post-grant challenge through a common object.

Future empirical work can take these predictions to data by linking patent text, prosecution histories, citation-source information, and post-grant challenge outcomes. The relevant empirical object is not only whether a patent is eventually challenged, but the full sequence of margins through which challenge risk can affect patenting: applicant-generated information, examiner-added information, claim breadth at application and grant, grant or abandonment, and post-grant challenge. A companion empirical analysis can use changes in post-grant challenge institutions, such as the introduction of PTAB review, to study whether downstream challenge risk feeds back into the production of patent breadth before grant.

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A Construction of Ex Ante Challenge Probabilities

This appendix provides additional detail on the construction of the challenge probabilities used in the applicant's accept-or-abandon decision. The applicant observes the post-examination state (m_2, R_2) and granted breadth b_E , but does not observe the challenger's privately known filing cost K_C . From the applicant's perspective, K_C is distributed according to the known distribution G_C .

For any possible challenger cost K_C , the challenger chooses equilibrium search

$$s_C^*(K_C),$$

which induces a distribution over challenger search outcomes X_C . Each realization of X_C produces a post-search record (m_3, R_3) and therefore a filing threshold

$$\hat{K}_C(b_E; m_3, R_3).$$

Thus, even for a fixed challenger cost, whether the challenger files remains probabilistic from the applicant's perspective because the outcome of challenger search is not yet known.

The type-specific filing propensity is

$$\pi_f(K_C; b_E, m_2, R_2) \equiv \Pr_{X_C | s_C^*(K_C)} \left(K_C \leq \hat{K}_C(b_E; m_3, R_3) \middle| m_2, R_2 \right).$$

The probability is taken over the search outcomes generated by $s_C^*(K_C)$, holding the possible challenger cost K_C fixed. Equivalently, $\pi_f(K_C; b_E, m_2, R_2)$ is the expected filing indicator over those search realizations.

The applicant does not know which challenger cost it faces. The ex ante filing probability therefore averages these type-specific filing propensities over the distribution G_C :

$$p_f(b_E; m_2, R_2) = \int \pi_f(K_C; b_E, m_2, R_2) dG_C(K_C).$$

Equivalently,

$$p_f(b_E; m_2, R_2) = \mathbb{E}_{K_C \sim G_C} [\pi_f(K_C; b_E, m_2, R_2)].$$

The construction therefore contains two nested sources of uncertainty. For each possible K_C , the

applicant first evaluates filing over the distribution of challenger search outcomes. It then averages that filing propensity over the possible values of K_C .

The same logic applies to the probability that the applicant both faces and loses a challenge. Conditional on filing, the expected probability of challenger success for a challenger with cost K_C is

$$\mathbb{E}_{X_C | s_C^*(K_C)} \left[\bar{p}_C(b_E; m_3, R_3) \middle| K_C \leq \hat{K}_C(b_E; m_3, R_3), m_2, R_2 \right].$$

The type-specific joint filing-and-loss probability is therefore

$$\pi_\ell(K_C; b_E, m_2, R_2) \equiv \pi_f(K_C; b_E, m_2, R_2) \mathbb{E}_{X_C | s_C^*(K_C)} \left[\bar{p}_C(b_E; m_3, R_3) \middle| K_C \leq \hat{K}_C(b_E; m_3, R_3), m_2, R_2 \right],$$

with $\pi_\ell(K_C; b_E, m_2, R_2) = 0$ whenever $\pi_f(K_C; b_E, m_2, R_2) = 0$. The applicant's ex ante probability of both facing and losing a challenge is then

$$p_\ell(b_E; m_2, R_2) = \int \pi_\ell(K_C; b_E, m_2, R_2) dG_C(K_C),$$

or equivalently,

$$p_\ell(b_E; m_2, R_2) = \mathbb{E}_{K_C \sim G_C} [\pi_\ell(K_C; b_E, m_2, R_2)].$$

This construction differs from simply evaluating

$$\mathbb{E} \left[G_C \left(\hat{K}_C(b_E; m_3, R_3) \right) \middle| m_2, R_2 \right].$$

The challenger's equilibrium search choice depends on its privately observed filing cost, so K_C affects the distribution of X_C and consequently the distribution of the filing threshold itself. The applicant must therefore evaluate filing behavior separately for each possible challenger cost and then integrate the resulting filing and loss propensities over G_C .

B Challenger Cost and the Equilibrium Filing Propensity

This subsection establishes that the equilibrium filing propensity is weakly decreasing in the challenger's realized filing cost, allowing the challenger to adjust its search choice.

For a fixed search choice s_C , define

$$W_C^{s_C}(K_C) \equiv \mathbb{E}_{X_C|s_C} \left[\max \left\{ 0, \hat{K}_C(b_E; m_3, R_3) - K_C \right\} \middle| m_2, R_2 \right] - c_C(s_C; \lambda_C).$$

For any realization of challenger search,

$$\max\{0, \hat{K}_C - K_C\}$$

is the maximum of two affine functions of K_C , and is therefore convex in K_C . Expectations preserve convexity, while $c_C(s_C; \lambda_C)$ is constant with respect to K_C for fixed s_C . Hence $W_C^{s_C}(K_C)$ is convex for every s_C .

The optimized challenger continuation value is

$$W_C(K_C) \equiv \max_{s_C \in S} W_C^{s_C}(K_C).$$

Because S is finite and the pointwise maximum of convex functions is convex, $W_C(K_C)$ is convex.

For a fixed s_C , increasing K_C lowers the realized payoff one-for-one whenever the challenger files and leaves the payoff unchanged whenever it does not file. Accordingly,

$$-\pi_f^{s_C}(K_C) \in \partial W_C^{s_C}(K_C),$$

where $\partial W_C^{s_C}(K_C)$ denotes the set of subgradients of the convex value function at K_C , and $\pi_f^{s_C}(K_C)$ is the filing propensity associated with search choice s_C . At points of differentiability this reduces to the usual slope relation; the subgradient notation accommodates filing thresholds at which the value function is kinked.

Let $s_C^*(K_C)$ be an equilibrium search choice. Because

$$W_C(K_C) = W_C^{s_C^*(K_C)}(K_C),$$

any subgradient of the active value function $W_C^{s_C^*(K_C)}$ is also a subgradient of W_C . Therefore,

$$-\pi_f(K_C) \in \partial W_C(K_C).$$

For a convex function, subgradients are weakly increasing in their argument. Hence, for any

$$K'_C > K_C,$$

and any

$$g \in \partial W_C(K_C), \quad g' \in \partial W_C(K'_C),$$

we have

$$g \leq g'.$$

Choosing

$$g = -\pi_f(K_C), \quad g' = -\pi_f(K'_C)$$

therefore gives

$$-\pi_f(K_C) \leq -\pi_f(K'_C),$$

or equivalently,

$$\pi_f(K'_C) \leq \pi_f(K_C).$$

Thus, higher realized challenger costs weakly reduce the equilibrium filing propensity even when the challenger adjusts its search choice optimally.